

Mayank Sewlia

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Skills

Languages: C++, Python, MATLAB, Simulink, ROS

Concepts: DRAKE, feedback control, real-time system, force control, formal methods, algorithms

Education

KTH Royal Institute of Technology – PhD in Electrical Engineering

Present

Technion Israel Institute of Technology – MSc in Aerospace Engineering

Aug 2020

Alliance University – B.Tech in Aerospace Engineering

Aug 2017

Selected Publications

Leader-Follower Cooperative Manipulation Under Spatio-Temporal Constraints,

IROS 2024

- Investigated object transport with unknown dynamics and task knowledge limited to one robot
- Designed stiffness-based and adaptive control laws for human-robot collaboration
- Conducted simulations using Python and the DRAKE toolbox

MAPS²: Multi-Robot Anytime Motion Planning under STL Specifications,

Under Review

- Developed motion planning algorithm for multi-robot systems under signal temporal logic
- Achieved probabilistic guarantees via distributed optimisation and sampling-based methods
- Created a [toolbox](#) in Python and tested on 3 mobile manipulator platforms

Cooperative Object Manipulation under STL Tasks and Uncertain Dynamics,

RAL 2022

- Worked on object transport under spatial and temporal constraints and gained first hardware experience
- Developed cooperative and decentralised methodologies for multi-arm manipulators
- Debugged control explosion and network issues using Python and HEBI Robotics API

Bearing-Based Formation Stabilization Using Event-Triggered Control,

IJRN 2024

- Researched event-triggered control to reduce communication and resource usage
- Provided theoretical convergence of the algorithm and avoided Zeno behaviour
- Implemented the algorithm in a distributed manner using MATLAB and Simulink

Professional Development

Supervisor for the following Master of Science Thesis

- Hampus Carlens: Manipulation on the move for pick and place tasks
- Rufus Wong: Optimizing Topologically Distinct Guesses for Mobile Manipulator Path Planning
- Sara Gomiero: Sampling-based synthesis of controllers for coupled agents under STL specifications

Teaching Assistant

- (Autumn 2022, Autumn 2024) EL1020 Automatic Control - Bachelor of Science course
- (Spring 2023) EL2520 Advanced Control Theory and Practice - Master of Science course

Internships

June 2016 – June 2017

- Designed an electrical harness connector system for a moon mission using finite element analysis (ISRO)
- Co-developed a CAD model of an aerofoil for supersonic wind tunnel testing (Additive Industries)
- Co-developed and performed stress analysis on a rear engine mount for the advanced light helicopter (HAL)

Leadership

- Coordinated with all 5 school committees at KTH as Council Coordinator for the PhD Chapter
- Served as Vice President of the KTH Rowing Board, managing marketing, travel, and introductory classes
- Co-founded Quasor Rocketry LLP, organising model chemical rocketry workshops for schools and colleges